

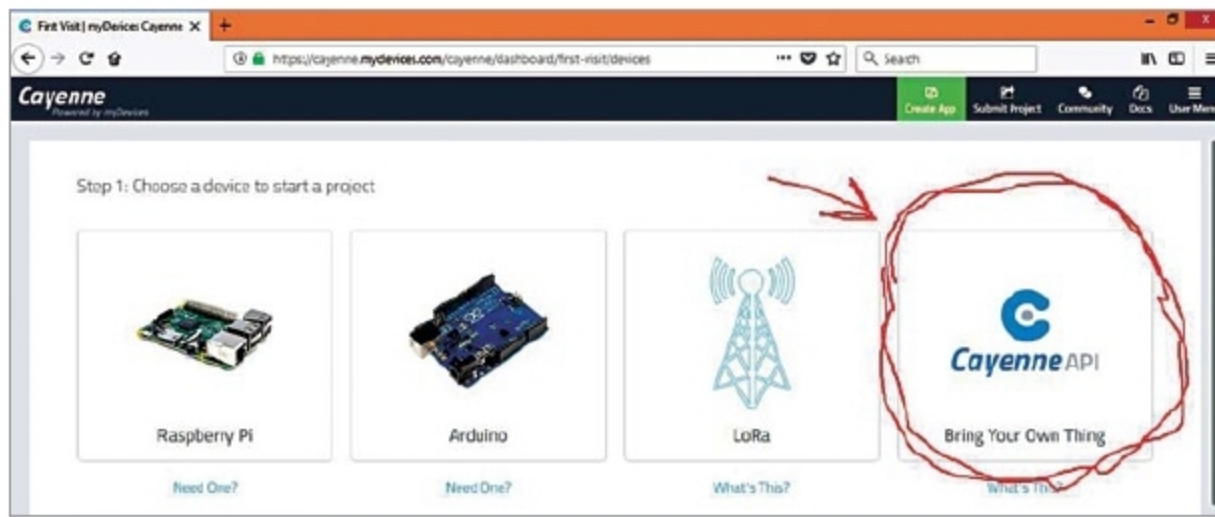


Fig. 4: Cayenne API

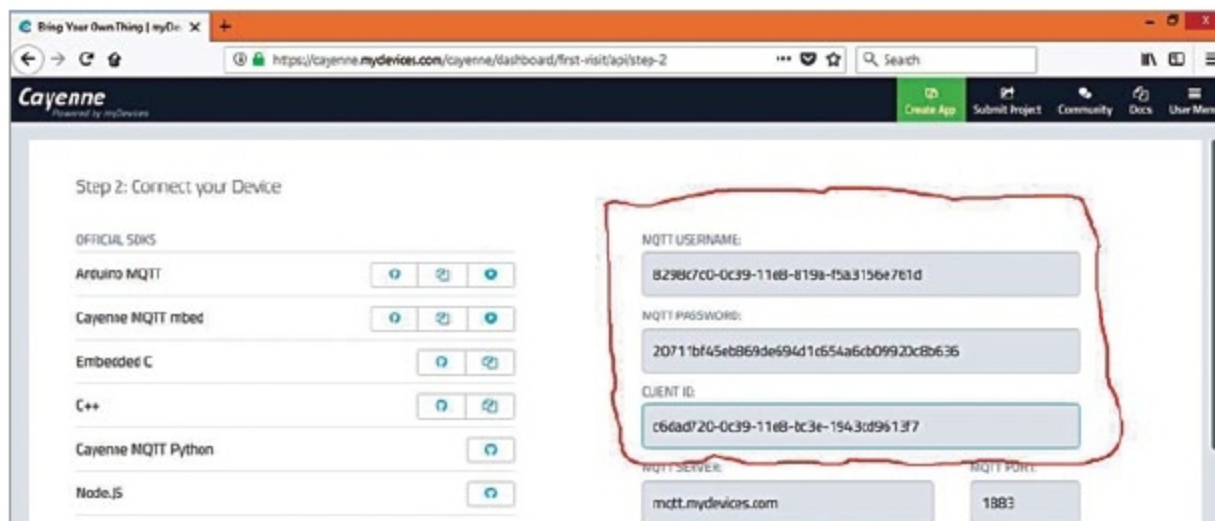


Fig. 5: MQTT credentials

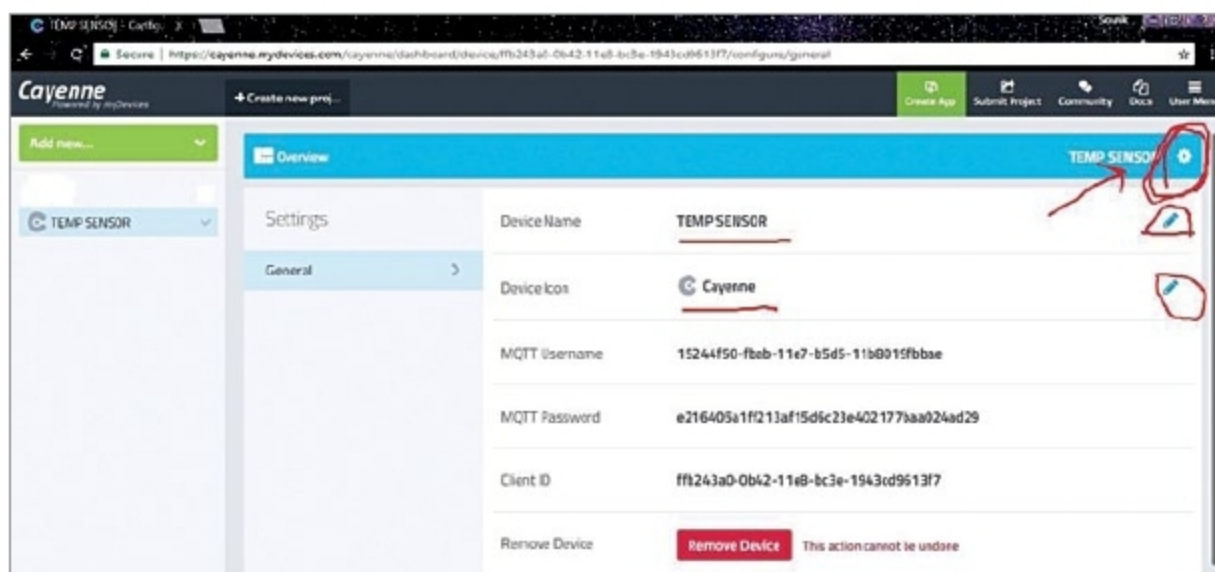


Fig. 6: Cayenne settings

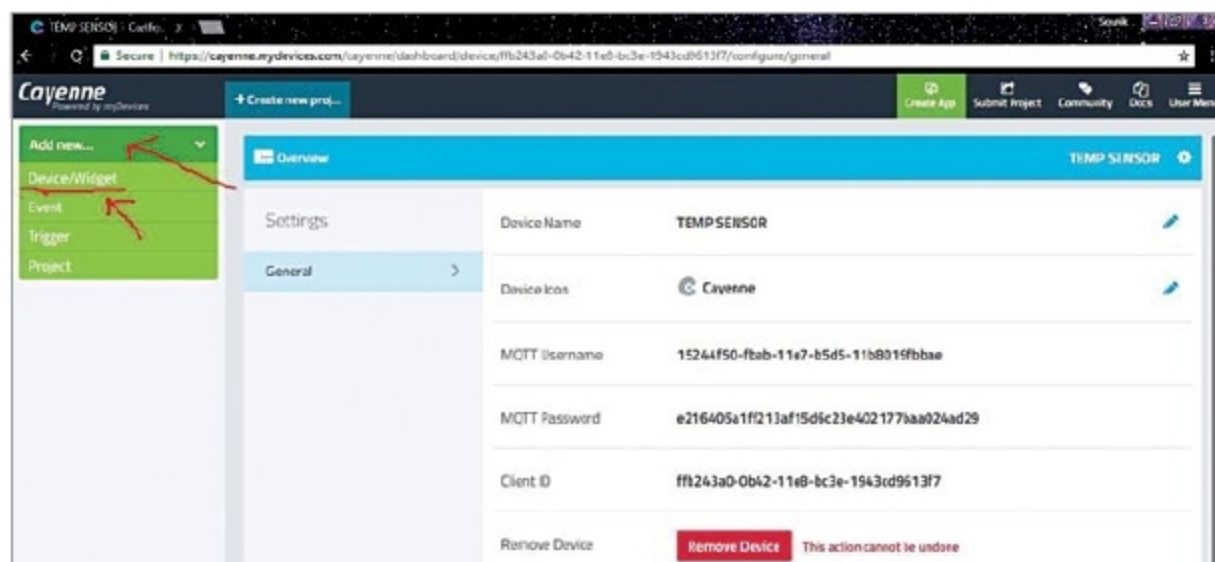


Fig. 7: Cayenne channel

3. Go to Tools and select Boards Manager.

4. Type esp8266 in search box and press Enter.

5. Install the latest version.

6. From Tools menu, select NodeMCU 1.0 (ESP-12E module) board.

7. Select Include Library from Sketch menu.

8. Download the library from <https://github.com/myDevicesIoT/Cayenne-MQTT-ESP>

9. Click on Add ZIP Library and add Cayenne-MQTT-ESP-master zip file, or directly copy the folder (Cayenne-MQTT-ESP-master) and paste it in Libraries folder of Arduino IDE.

Software

The source code (cayenne_sensor.ino) to program NodeMCU is written in Arduino programming language. ESP8266-12E is programmed using Arduino IDE software. Open Arduino IDE and select the correct board from Tools→Board menu in Arduino IDE.

The source code starts with `#define CAYENNE_PRINT Serial` and `#include <CayenneMQTTESP8266.h>`

Here, CayenneMQTTESP8266.h header file is used to interface NodeMCU with <https://cayenne.mydevices.com>

Cayenne.begin() is used to connect Wi-Fi and Cayenne dashboard with NodeMCU.

Cayenne_out_default() and Cayenne.virtualWrite(1,a) are used for writing sensor data ('a') on channel number 1.

Serial.begin(9600) function initialises the serial port with a baud rate of 9600.

Enter your Wi-Fi name and password in the source code, as indicated by xxxxx below. Also provide other details in the source code, such as username, password and client ID obtained from Cayenne as explained in the next section.

```
#define CAYENNE_PRINT Serial
#include <CayenneMQTTESP8266.h>
// WiFi network info.
char ssid[] = "xxxxx"; //put your own ssid
char wifiPassword[] = "xxxxx"; //put your
    wifi password
// Cayenne authentication info. This should
    be obtained from the Cayenne Dashboard.
```